

B. Tech Degree V Semester (Supplementary) Examination July 2009

ME 502 METROLOGY AND MACHINE TOOLS

(2006 Scheme)

Time : 3 Hours

Maximum Marks : 100

(Sketches are to be included wherever necessary)

PART – A

(Answer ALL questions)

(8 x 5 = 40)

- I. (a) Differentiate between Gross Errors and Random Errors.
(b) What do you understand by line and end measurements? Compare their relative characteristics.
(c) Explain the working of a Clinometer.
(d) With the help of a suitable sketch explain the optical system of an Optical Projector.
(e) List the parameters needed to specify a Lathe Completely.
(f) Differentiate between Planer and Shaper.
(g) Distinguish between Wheel Truing and Wheel Dressing.
(h) Distinguish between NC and CNC.

PART – B

(4 x 15 = 60)

- II. (a) Define and distinguish between the following terms.
(i) Accuracy (ii) Precision
(iii) Sensitivity (iv) Readability (8)
(b) Explain the term Cosine Error with a suitable example. (7)
- OR**
- III. (a) Differentiate between limits and tolerances. (5)
(b) Write short notes on :
(i) Plain Ring Gauges (ii) Snap Gauges (10)
- IV. (a) Distinguish between Sine Table and Sine Centre. (5)
(b) With the help of suitable sketches explain any two uses of angle gauges in measurements. (10)
- OR**
- V. With the help of suitable sketches explain the following :
(i) Optical Flat (ii) Tool Maker's Microscope
(iii) Spirit Level (15)
- VI. With the help of a suitable sketches explain in detail the Tool Nomenclature for a single point cutting Tool used in a Lathe. (15)
- OR**
- VII. With the help of suitable sketch explain the Quick-Return Mechanism and ram positioning equipment used in a shaper. (15)
- VIII. (a) Differentiate between Cylindrical Grinding and Centreless Grinding. (5)
(b) What do you understand by Honing? Explain in detail. (10)
- OR**
- IX. Write short notes on :
(i) CAPP (ii) APT (iii) CNC (15)

